

Hotel Booking Cancellation Analysis

1. Background

The hospitality industry operates in an environment of persistent demand uncertainty, and hotel booking cancellations represent one of its most consequential operational challenges. As online travel platforms have normalized flexible booking policies that allow guests to cancel with little or no financial penalty, the rate of cancellations across the industry has risen steadily. For hotel revenue managers, this creates a recurring problem: rooms that appear occupied on paper may ultimately go empty, while last-minute vacancies are difficult to refill at competitive rates.

This project addresses the problem as a supervised binary classification task. The target variable is whether a given booking will be canceled (`is_canceled = 1`) or not (`is_canceled = 0`). By training machine learning models on historical booking data, we aim to identify the characteristics that most reliably distinguish canceling guests from non-canceling guests, and to produce predictions accurate enough to support proactive revenue management decisions at the property level.

The project pursues three complementary research questions:

- (1) Which booking characteristics significantly affect the probability of cancellation?
- (2) Which customer segments carry the highest cancellation risk?
- (3) Can cancellations be predicted with sufficient accuracy for operational deployment? These questions are addressed using logistic regression, K-Means clustering, and XGBoost respectively, together covering the spectrum from interpretability to predictive performance.

2. Data Selection

2.1 Dataset Description

The dataset used in this project is the Hotel Booking Demand dataset published by Antonio, Almeida, and Nunes (2019) and made available through Kaggle. It contains 119,390 booking records from two hotels — a city hotel and a resort hotel — covering multiple years of reservations. Each row represents a single booking and includes 36 original attributes spanning booking logistics (lead time, deposit type, market segment, distribution channel), booking history (previous cancellations, previous bookings not canceled), stay details (hotel type, ADR, number of nights), and guest profile (repeat guest status, country, number of special requests, party composition). The binary target variable, `is_canceled`, takes the value 1 for canceled bookings and 0 for not canceled bookings. Of the 119,390 bookings, approximately 37% of hotel bookings are canceled which is equivalent to roughly 1 out of every 3 bookings ending in cancellation.

This indicates that cancellation is a frequent and economically meaningful outcome rather than a rare event.

2.2 Data Cleaning

1. **Deduplication.** Exact duplicate rows were removed first to prevent any individual booking from having disproportionate influence on model parameters.
2. **Leakage column removal.** The columns `reservation_status` and `reservation_status_date` were dropped immediately, as both directly encoded the cancellation outcome and would produce trivially inflated model accuracy without any predictive validity.
3. **Type coercion.** Eleven columns were explicitly cast to numeric using `pd.to_numeric` with `errors='coerce'`, converting any malformed values to NaN rather than causing downstream errors.
4. **Missing value treatment:** The `children` column was filled with 0 (no children assumed when absent). The `country` column was filled with 'Unknown'. For agents and companies, binary indicator flags were created before filling missing values with 0, preserving the distinction between a genuine zero and a missing entry.
5. **Zero-guest records:** 180 bookings (only 0.15% of the dataset) were found to have zero total guests (combination of children, adults...) — a logistically impossible value. As mentioned in the presentation, these were flagged as data quality anomalies and retained in the dataset, noted as a limitation rather than removed, given their negligible share of the data.
6. **ADR floor clip.** Negative ADR values were clipped to 0 using a fixed domain rule. This is a deterministic correction, not a learned statistic, and introduces no leakage.
7. **Feature engineering.** Two composite features were constructed: `total_nights` (sum of weekday and weekend nights) and `total_guests` (sum of adults, children, and babies). Both are arithmetic derivations of existing columns.
8. **Post-split ADR engineering.** Two additional features — `adr_capped` (ADR clipped at the 99th percentile) and `booking_value` (`adr_capped` $\times$ `total_nights`) — were computed after the train/test split. The 99th percentile cap was calculated exclusively on the training set and applied to the test set, ensuring that test data does not influence the cap value. Raw `adr` and `total_nights` were subsequently dropped.

2.3 Feature Statistics (EDA)

After the data cleaning, the dataset retains 119,390 rows. The table below summarizes the key features used across all three models.

Feature	Type	Description
<code>lead_time</code>	Numeric	Days between booking and arrival. Canceled median: 120 days; kept: 45 days.

deposit_type	Categorical	No Deposit (87.6%), Non Refund (12.2%), Refundable (0.1%).
is_repeated_guest	Binary	1 if a guest has stayed before. Repeat guests cancel at 14.6% vs 37.8%.
previous_cancellations	Numeric	Count of prior cancellations. Guests with history cancel at 91.6%.
distribution_channel	Categorical	TA/TO (79%), Direct (12%), Corporate (6%), GDS (2%).
total_of_special_requests	Numeric	0–5 requests. Cancel rate falls from 48% (0 requests) to ~5% (5+).
adr_capped	Numeric	Average daily rate capped at training-set 99th percentile.
booking_value	Numeric	$\text{adr_capped} \times \text{total_nights}$. Total financial commitment proxy.
hotel	Categorical	City Hotel (66%): 42% cancel rate. Resort Hotel (34%): 28%.

3. Model Section

3.1 Model 1— Logistic Regression (Yuhan Luan)

3.1 a) Motivation

Hotel booking cancellations follow patterns rather than occurring at random — they tend to cluster around certain booking behaviors, channels, and reservation characteristics. We expected that bookings made far in advance, through higher-risk channels, or with fewer commitment signals would cancel at higher rates. Logistic regression was chosen because the outcome variable is `is_canceled` is binary (1 = canceled, 0 = not canceled), and beyond just predicting probabilities, the model yields interpretable coefficients that tell us who cancels and why — which matters as much as prediction accuracy for this analysis.

3.1 b) Feature Selection

15 features were selected based on domain knowledge and EDA findings, including numeric variables (`lead_time`, `previous_cancellations`, `previous_bookings_not_canceled`, `total_of_special_requests`, `adr`, `total_nights`, `adults`, `children`, `babies`) and categorical variables (`hotel`, `market_segment`, `distribution_channel`, `deposit_type`, `customer_type`, `is_repeated_guest`).

3.1 c) Data Split and Preprocessing

The data was split into 75% training (89,542 observations) and 25% test (29,848 observations) using stratified sampling (`stratify=y`, `random_state=42`), preserving the 37.04% cancellation rate in both sets.

Preprocessing was implemented inside a scikit-learn Pipeline with ColumnTransformer to prevent data leakage across cross-validation folds: numeric features were imputed with the median and standardized with StandardScaler; categorical features were imputed with the most frequent value and encoded with OneHotEncoder (`drop='first'` to avoid multicollinearity).

3.1 d) Hyperparameter

The key hyperparameter choice is `penalty=None` (no regularization). Since Q1's primary goal is *inferential* — identifying which features significantly drive cancellations — unbiased MLE coefficient estimates are preferred over regularized shrinkage. `max_iter` was set to 2000 to ensure convergence on the high-dimensional feature space. Statistical significance was verified by re-fitting the model using *statsmodels.Logit*, which yields z-statistics and p-values. The model achieves a McFadden's Pseudo R^2 of 0.3173, indicating good fit.

3.1 e) Evaluation Metrics

Five metrics are reported: Accuracy, Precision, Recall, F1-Score, and ROC-AUC. For hotel operations, Recall is arguably the most important — missing an actual cancellation (false negative) is typically more costly than a false alarm.

3.2. Model 2 — K-Means Clustering (Jiaru Li)

3.2 a) Motivation

Predictive classification models can identify whether a booking is likely to be canceled; however, they do not reveal the underlying behavioral profiles. To move beyond simple risk scores and enable targeted, segment-specific interventions, unsupervised clustering was applied to segment hotel guests into distinct risk groups. This section addresses which customer profiles carry the highest cancellation risk and what their defining characteristics are.

3.2 b) Feature Selection & Data Preprocessing

Five features were selected to capture both the financial value and behavioral characteristics of each booking: `lead_time`, `adr_capped`, `total_nights`, `booking_value`, and `total_of_special_requests`. All five features were standardized using StandardScaler before fitting to ensure that no single variable dominates the clustering objective due to scale differences.

3.2 c) Select the number of clusters

The optimal number of clusters K was determined using two complementary approaches:

- The Elbow Method (Within-Cluster Sum of Squares, WCSS)
- Silhouette Scores.

Table 1. Silhouette Scores for K = 2 to 5

K = 2	K = 3	K = 4	K = 5
0.331	0.238	0.263	0.268

Although $K = 2$ yields the highest silhouette score (0.331), it provides insufficient granularity for practical hotel management. The resulting segments are too broad to support differentiated policy decisions. The difference in silhouette score between $K = 4$ (0.264) and $K = 5$ (0.268) is less than 0.005, which falls within the range of sampling variability and does not constitute a meaningful improvement in cluster quality. Furthermore, the Elbow Method identifies a clear inflection point at $K = 4$. Each additional cluster beyond four would require a distinct marketing strategy and operational workflow, without a commensurate gain in statistical or practical coherence. Accordingly, $K = 4$ was selected as the optimal number of clusters.

3.2 d) Algorithm Configuration

K-Means clustering with the k-means++ initialization strategy was applied using `random_state = 42` for reproducibility. The k-means++ initialization selects initial centroids in a probabilistically spread manner, reducing sensitivity to random starting points and improving convergence stability.

3.3. Model 3 — XGBoost (Xinru Tang)

3.3 a) Motivation

Hotel booking cancellations lead to significant revenue loss and operational inefficiency, with over one in three bookings in this dataset never resulting in an actual stay. XGBoost (Extreme Gradient Boosting) was selected as the primary predictive model to address this problem. It is a gradient boosting ensemble method that builds decision trees sequentially, where each new tree focuses on correcting the errors made by previous trees. XGBoost is widely regarded as one of the most effective algorithms for structured tabular data due to its ability to capture complex non-linear relationships, handle mixed feature types natively, and produce built-in feature importance scores that support interpretability. The model pursues two goals: predicting whether a guest will cancel their reservation, and identifying the key booking characteristics that drive that outcome.

3.3 b) Feature Selection & Engineering

Five columns were removed prior to modeling: `reservation_status` and `reservation_status_date` (direct leakage of the target variable) and four personal info-related columns (`name`, `email`, `phone-number`, `credit_card`) which carry no generalizable predictive signal.

The country feature originally contained over 170 unique values; to reduce sparsity, only the top 10 most frequent countries were retained and all remaining values grouped into an "Other" category.

Several features were also engineered to capture higher-level booking characteristics not directly available in the raw data. `total_nights` aggregates weekday and weekend nights into a single length-of-stay measure, which may correlate with booking commitment. `total_guests` sums adults, children, and babies into a unified party size variable, as larger parties may exhibit different cancellation behavior. Binary indicator flags `agent_missing` and `company_missing` were created before imputation to preserve the distinction between a genuine zero agent or company code and a missing entry, which may reflect different booking pathway characteristics. After cleaning, engineering, and cardinality reduction, the feature matrix contains 119,390 rows and 33 features (23 numeric, 10 categorical).

3.3 c) Preprocessing Pipeline

A `ColumnTransformer` handles numeric and categorical features separately. Numeric features were processed through a median imputer followed by `StandardScaler` normalization.

Categorical features were processed through a mode imputer followed by `OneHotEncoder` with `handle_unknown="ignore"` to gracefully handle unseen categories at inference time.

3.3 d) Train-Test Split

The dataset was split 80/20 (Train: 95,512 rows; Test: 23,878 rows) with stratification on the target variable to preserve the cancellation rate distribution in both sets.

3.3 e) Hyperparameter Tuning

Hyperparameter tuning was conducted via `GridSearchCV` with 5-fold stratified cross-validation (`StratifiedKFold`, `shuffle=True`, `random_state=42`) across 32 parameter combinations, producing 160 total fits. The search grid covered: `n_estimators` $\in \{200, 300\}$, `max_depth` $\in \{4, 6\}$, `learning_rate` $\in \{0.05, 0.1\}$, `subsample` $\in \{0.8, 1.0\}$, and `colsample_bytree` $\in \{0.8, 1.0\}$. Models were scored on ROC-AUC during cross-validation.

3.3 f) Evaluation Metrics

Models were evaluated on accuracy, precision, recall, F1-score, and ROC-AUC on the held-out test set. ROC-AUC was the primary criterion during cross-validation as it is threshold-independent and more robust than accuracy under class imbalance. Precision captures the proportion of predicted cancellations that are genuine, recall captures the proportion of actual cancellations detected, and F1-score balances both.

4. Empirical Results

4.1. Logistic Regression Results

4.1 a) Test Set Performance

Metric	Value
Accuracy	0.8024
Precision	0.8584
Recall	0.5587
F1-Score	0.6769
ROC-AUC	0.8284

The model achieves an ROC-AUC of 0.8284, demonstrating strong discriminative ability. Accuracy of 80.24% is solid; however, Recall of 55.87% indicates that approximately 44% of actual cancellations are missed at the default 0.5 threshold — a typical trade-off when prioritizing precision. The high Precision (85.84%) means that flagged bookings are correct nearly 86% of the time.

4.2 b) Coefficient Estimates and Odds Ratios

The tables below present significant features ($p < 0.05$) sorted by effect size.

Table 2: Features that increase cancellation probability

Feature	Coef.	Odds Ratio	Interpretation
deposit_type_Non Refund	+5.721	305.1	Strongest predictor; non-refundable deposit strongly associated with cancellation
previous_cancellations	+2.520	12.4	Each prior cancellation multiplies future cancellation odds by 12.4×
customer_type_Transient	+0.834	2.30	Transient customers more likely to cancel
market_segment_Online TA	+0.469	1.60	Online OTA bookings have higher cancellation odds

lead_time	+0.409	1.51	Longer advance booking window increases cancellation risk
-----------	--------	------	---

Table 3: Features that decrease cancellation probability

Feature	Coef.	Odds Ratio	Interpretation
previous_bookings_not_canceled	−0.983	0.374	Completed stays strongly reduce cancellation risk
market_segment_Offline TA/TO	−0.870	0.419	Offline bookings are significantly more stable
total_of_special_requests	−0.589	0.555	Each special request reduces cancellation odds by ~44.5%
distribution_channel_Direct	−0.502	0.605	Direct bookings show higher commitment
hotel_Resort Hotel	−0.160	0.852	Resort bookings ~15% less likely to cancel than city hotels

4.3 c) Key Insights

(1) Deposit Type Paradox: The Non-Refund deposit carries an odds ratio of 305.1, by far the largest effect in the model. A likely explanation is that non-refundable bookings are often made speculatively at low prices far in advance; when circumstances change, guests cancel despite forfeiting the deposit. This suggests that non-refundable pricing strategies may not deter cancellations as intended.

(2) Booking History as the Most Actionable Signal: previous_cancellations (OR = 12.4) and previous_bookings_not_canceled (OR = 0.374) together suggest that a guest's historical booking behavior is among the most valuable features for individual risk scoring.

(3) Special Requests as a Commitment Proxy: Each additional special request reduces cancellation odds by approximately 44.5%, serving as a behavioral signal of guest investment and commitment to the stay.

4.2 K-Means Results

The K-Means algorithm identified four distinct customer segments. The following Table 2 summarizes the mean feature values, cancellation rate, and population share for each cluster.

Table 2. Cluster Profiles — Mean Feature Values and Cancellation Rates

Cluster	Lead Time (days)	ADR (€)	Total Nights	Booking Value (€)	Special Requests	Cancel Rate	Population Share
0 – Budget Travelers	44.9	82.97	2.54	207.26	0.21	32.5%	45.1%
1 – Premium Business	71.2	127.30	3.16	380.43	1.54	23.0%	23.8%
2 – Early Bird Vacationers	121.6	149.99	8.01	1,082.05	0.63	41.9%	10.7%
3 – Speculative Planners	263.8	86.77	3.31	278.29	0.20	60.9%	20.4%

The four segments can be characterized as follows:

Cluster 0: Budget Travelers (Low Risk, 45.1%). This is the hotel's core customer base, comprising nearly half of all bookings. Guests in this segment book approximately 45 days in advance, stay for short trips (averaging 2.5 nights), and pay a relatively modest average daily rate of €83. They generate a few special requests (mean = 0.21), suggesting limited pre-arrival planning. Their cancellation rate of 32.5% is below the dataset average of 37.0%, making them a predictable and manageable segment. Only 4.9% have a prior cancellation history, indicating generally reliable booking behavior.

Cluster 1: Premium Business Travelers (Lowest Risk, 23.8%). This segment represents the hotel's highest-value guests. They reserve approximately 71 days ahead with a significantly higher ADR of €127 and make an average of 1.54 special requests, the most of any segment, indicating thorough pre-arrival planning and a strong commitment to their stay. Their cancellation rate of 23.0% is the lowest across all clusters, and only 2.2% have a prior cancellation history. These guests generate a mean booking value of €380, making their retention strategically important.

Cluster 2: Early Bird Vacationers (Moderate-High Risk, 10.7%). Guests in this segment book approximately 122 days (four months) in advance, pay the highest ADR among all clusters (€150), and stay for the longest trips (averaging 8 nights), resulting in a substantially higher mean booking value of €1,082. Despite this financial commitment, their cancellation rate reaches 41.9%, which is above the dataset average. The extended planning horizon offers a broad timeframe during which plans can change, as guests might find better rates, encounter scheduling conflicts, or lose interest. This segment represents a moderate-to-high-risk group that merits proactive mid-lead window engagement.

Cluster 3: Speculative Planners (Highest Risk, 20.4%). This segment presents the highest operational risk to hotel revenue. Guests typically book about 264 days, or roughly eight months, in advance at a low ADR of €87, with minimal special requests, averaging 0.20. This indicates that bookings are often tentative, serving as placeholders without a firm intention to stay. Their cancellation rate of 60.9% is nearly double the overall dataset average, and 28.1% have a prior cancellation history, confirming a recurring behavioral pattern. Despite representing only 20.4% of bookings, this segment is the primary driver of revenue uncertainty and inventory waste.

4.3 XGBoosting Results

4.3 a) Test set performance

Metric	Value
Accuracy	0.8820
Precision	0.8616
Recall	0.8120
F1-Score	0.8360
ROC-AUC	0.9558

The model achieves an AUC of 0.9558, indicating excellent discriminative ability. With an accuracy of 88.2%, the model correctly classifies nearly 9 out of 10 bookings. The precision of 86.2% means that when the model predicts a cancellation, it is correct about 86% of the time. The recall of 81.2% means the model catches approximately 4 out of 5 actual cancellations.

4.3 b) Confusion Matrix

	Predicted: Not Cancel	Predicted: Cancel
Actual: Not Cancel	13,879 (TN)	1,154 (FP)
Actual: Cancel	1,663 (FN)	7,182 (TP)

The confusion matrix shows 13,879 true negatives and 7,182 true positives, meaning the model correctly classifies the majority of both non-canceled and canceled bookings. The 1,154 false positives represent non-cancellations incorrectly flagged as high-risk — a low-cost error in practice, as the consequence is simply an unnecessary reminder or outreach to a guest who would have shown up anyway. The more meaningful failure mode is the 1,663 false negatives: actual

cancellations the model missed entirely, representing lost opportunities for proactive intervention and the primary area for future improvement.

4.3 c) ROC Curve

The ROC curve shows the model achieves an AUC of 0.96 (rounded), with the curve hugging the upper-left corner. This confirms that the model can reliably distinguish between cancelled and non-cancelled bookings across all threshold settings.

5. Conclusion: From Prediction to Action

5.1 How the Model Works in Practice

When a new booking comes in, the trained XGBoost model takes the booking's features (deposit type, lead time, market segment, previous cancellation history, etc.) as input and outputs a cancellation probability between 0 and 1. Bookings with a high predicted probability represent the guests most likely to cancel.

5.2 Targeted Strategies by Risk Driver

Based on the high-risk guest profile identified above, we recommend the following targeted strategies:

Risk Driver	Finding	Recommended Action
Deposit type = Non Refund	99.4% cancellation rate; dominates model predictions (57.5% importance)	Review deposit policies for high-risk profiles. Consider requiring partial prepayment for bookings that combine multiple risk factors (long lead time + OTA + no special requests).
Lead time > 180 days	57.0% cancellation rate, 6x higher than last-week bookings (9.6%)	Implement automated confirmation reminders at 30-day and 7-day marks for any booking with lead time exceeding 90 days.
Market segment = Online TA	36.7% cancellation rate, more than double the Direct rate (15.3%)	Drive direct bookings through loyalty programs, best-price guarantees, and exclusive perks. Negotiate stricter cancellation terms with OTA partners.
Has previous cancellations	91.6% cancellation rate vs 33.9% for first-time guests	Flag repeat cancellers in the reservation system and automatically apply stricter conditions (full prepayment or non-refundable rates) for their future bookings.